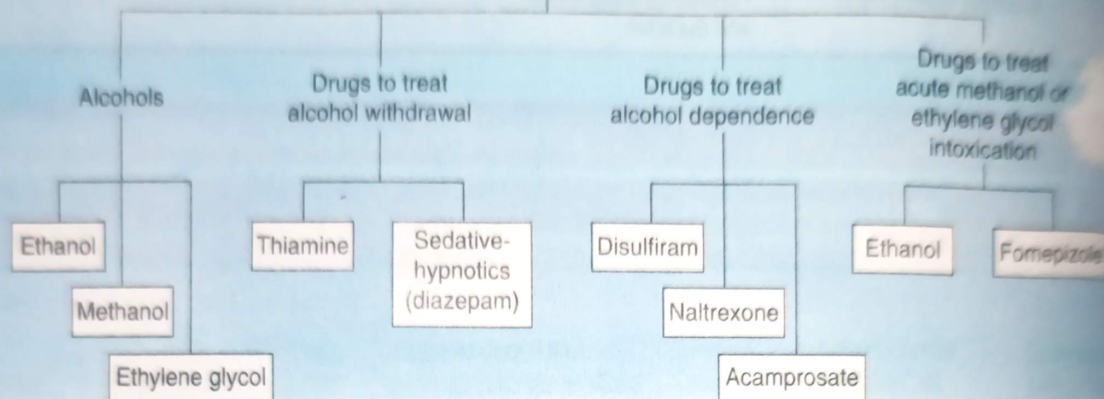


Ethanol, a sedative-hypnotic drug, is the most important alcohol of pharmacologic interest. It has few medical applications, but its abuse causes major medical and socioeconomic problems. A very large fraction of fatal automobile accidents are associated with "driving under the influence" (DUI) of ethanol.

Other alcohols of toxicologic importance include ethylene glycol. Several important drugs discussed later are used to prevent the potentially life-threatening withdrawal syndrome, to treat chronic alcoholism, and to treat acute methanol and ethylene glycol poisoning.

#### Clinically important alcohols and their antagonists



## ETHANOL

### A. Pharmacokinetics

After ingestion, ethanol is rapidly and completely absorbed; the drug is then distributed to most body tissues, and its volume of distribution is equivalent to that of total body water (0.5–0.7 L/kg). Two enzyme systems metabolize ethanol to acetaldehyde (Figure 23–1) and because these enzymes are essentially saturated by the large doses of ethanol commonly consumed, elimination of ethanol follows zero-order kinetics.

**1. Alcohol dehydrogenase**—This family of cytosolic,  $\text{NAD}^+$ -dependent enzymes, found mainly in the liver and gut, accounts for the metabolism of low to moderate doses of ethanol. Because of the limited supply of the coenzyme  $\text{NAD}^+$ , the reaction has *zero-order kinetics*, resulting in a fixed capacity for ethanol metabolism of 7–10 g/h. Gastrointestinal metabolism of ethanol is lower in

women than in men. Genetic variation in alcohol dehydrogenase (ADH) affects the rate of ethanol metabolism and with alcohol-use disorders.

**2. Microsomal ethanol-oxidizing system**—At blood alcohol levels higher than 100 mg/dL, the liver microsomal mixed-function oxidase system that catalyzes most phase I drug-metabolisms (see Chapter 4) contributes significantly to ethanol metabolism (Figure 23–1). Chronic ethanol consumption induces cytochrome P450 enzyme synthesis and microsomal ethanol-oxidizing (MEOS) activity; this is partially responsible for the development of tolerance to ethanol. The primary isoform of cytochrome P450 induced by ethanol—2E1 (see Table 4–3)—converts ethanol to a hepatotoxic metabolite.

Acetaldehyde formed from the oxidation of ethanol by ADH or MEOS is rapidly metabolized to acetate by aldehyde dehydrogenase, a mitochondrial enzyme found in the

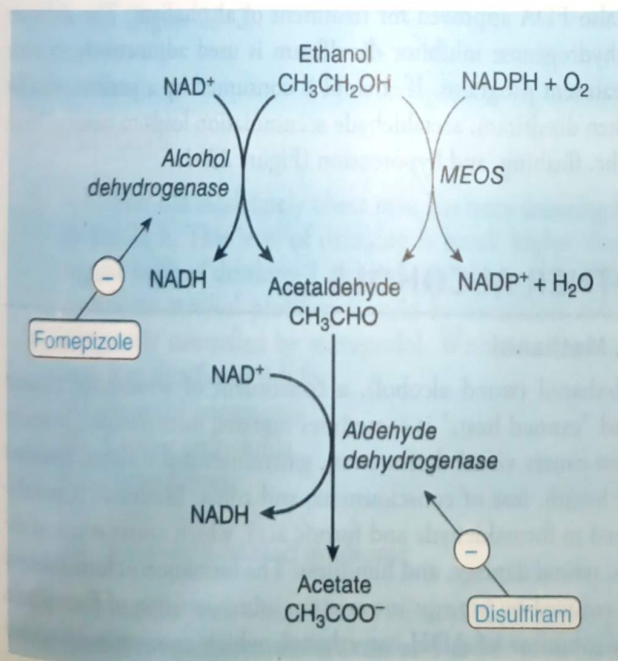
## High-Yield Terms to Learn

|                                    |                                                                                                                                                                                                                |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alcohol abuse</b>               | An alcohol-use disorder characterized by compulsive use of ethanol in dangerous situations (eg, driving, combined with other CNS depressants) or despite adverse consequences directly related to the drinking |
| <b>Alcohol dependence</b>          | An alcohol-use disorder characterized by alcohol abuse plus physical dependence on ethanol                                                                                                                     |
| <b>Alcohol withdrawal syndrome</b> | The characteristic syndrome of insomnia, tremor, agitation, seizures, and autonomic instability engendered by deprivation in an individual who is physically dependent on ethanol                              |
| <b>Delirium tremens (DTs)</b>      | Severe form of alcohol withdrawal whose main symptoms are sweating, tremor, confusion, and hallucinations                                                                                                      |
| <b>Fetal alcohol syndrome</b>      | A syndrome of craniofacial dysmorphia, heart defects, and mental retardation (intellectual disability) caused by the teratogenic effects of ethanol consumption during pregnancy                               |
| <b>Wernicke-Korsakoff syndrome</b> | A syndrome of ataxia, confusion, and paralysis of the extraocular muscles that is associated with chronic alcoholism and thiamine deficiency                                                                   |

many other tissues. Aldehyde dehydrogenase is inhibited by **disulfiram** and other drugs, including **metronidazole**, **oral hypoglycemics**, and some **cephalosporins**. Some individuals, primarily of East Asian descent, have genetic deficiency of aldehyde dehydrogenase. After consumption of even small quantities of ethanol, these individuals experience nausea and a flushing reaction from accumulation of acetaldehyde.

### B. Acute Effects

**1. CNS**—The major acute effects of ethanol on the CNS are sedation, loss of inhibition, impaired judgment, slurred speech, and ataxia. In nontolerant persons, impairment of driving ability is



**FIGURE 23-1** Metabolism of ethanol by alcohol dehydrogenase (ADH) and the microsomal ethanol-oxidizing system (MEOS). Alcohol dehydrogenase and aldehyde dehydrogenase are inhibited by fomepizole and disulfiram, respectively. (Reproduced with permission from Katzung BG, Vanderah TW: *Basic & Clinical Pharmacology*, 15th ed. New York, NY: McGraw Hill; 2011)

thought to occur at ethanol blood levels between 60 and 80 mg/dL. Blood levels of 120 to 160 mg/dL are usually associated with gross drunkenness. Levels greater than 300 mg/dL may lead to loss of consciousness, anesthesia, and coma sometimes with fatal respiratory and cardiovascular depression. Blood levels higher than 500 mg/dL are usually lethal. Individuals with alcohol dependence who are tolerant to the effects of ethanol can function almost normally at much higher blood concentrations than occasional drinkers, although the acutely lethal concentration is not elevated much. Additive CNS depression occurs with concomitant ingestion of ethanol and a wide variety of CNS depressants, including sedative-hypnotics, opioid agonists, and many drugs that block muscarinic and  $H_1$  histamine receptors. The molecular mechanisms underlying the complex CNS effects of ethanol are not fully understood. Specific receptors for ethanol have not been identified. Rather, ethanol appears to modulate the function of a number of signaling molecules. It facilitates the action of GABA at  $GABA_A$  receptors, inhibits the ability of glutamate to activate NMDA (*N*-methyl-D-aspartate) receptors, and modifies the activities of adenylyl cyclase, phospholipase C, and ion channels.

**2. Other organ systems**—Ethanol, even at relatively low blood concentrations, significantly depresses the heart. Vascular smooth muscle is relaxed, which leads to vasodilation, which in cold environments causes marked hypothermia.

### C. Chronic Effects

**1. Tolerance and dependence**—Tolerance occurs mainly as a result of CNS adaptation and, to a lesser extent, by an increase in the rate of ethanol metabolism. There is cross-tolerance to sedative-hypnotic drugs that facilitate GABA activity (eg, benzodiazepines and barbiturates). Both psychological and physical dependence are marked and constitute *alcohol-use disorder* or *alcoholism*.

**2. Liver**—Liver disease is the most common medical complication of chronic alcohol abuse. Progressive loss of liver function occurs, with reversible fatty liver progressing to irreversible hepatitis, cirrhosis, and liver failure. Hepatic dysfunction is often more



in women than in men and in both men and women infected with hepatitis B or C virus.

**3. Gastrointestinal system**—Irritation, inflammation, bleeding, and scarring of the gut wall occur after chronic heavy use of ethanol and may cause absorption defects and exacerbate nutritional deficiencies. Chronic alcohol abuse greatly increases the risk of pancreatitis.

**4. Neurologic**—Peripheral neuropathy is the most common neurologic abnormality in alcohol abuse. More rarely, thiamine deficiency, along with alcohol abuse, leads to **Wernicke-Korsakoff syndrome**, which is characterized by ataxia, confusion, and paralysis of the extraocular muscles. Prompt treatment with parenteral thiamine is essential to prevent a permanent memory disorder known as Korsakoff psychosis.

**5. Endocrine system**—Gynecomastia, testicular atrophy, and salt retention can occur, partly because of altered steroid metabolism in the cirrhotic liver.

**6. Cardiovascular system**—Excessive chronic ethanol use is associated with an increased incidence of hypertension, anemia, and dilated cardiomyopathy. Acute drinking for several days ("binge" drinking) can cause arrhythmias. However, the ingestion of modest quantities of ethanol (10–15 g/day) raises serum levels of high-density lipoprotein (HDL) cholesterol and may protect against coronary heart disease.

**7. Fetal alcohol syndrome**—Ethanol use in pregnancy is associated with teratogenic effects that include mental retardation (most common), growth deficiencies, microcephaly, and a characteristic underdevelopment of the midface region.

**8. Neoplasia**—Ethanol is not a primary carcinogen, but its chronic use is associated with an increased incidence of neoplastic diseases in the gastrointestinal tract and a small increase in the risk of breast cancer.

**9. Immune system**—Chronic alcohol abuse has complex effects on immune functions because it enhances inflammation in the liver and pancreas and inhibits immune function in other tissues. Heavy use predisposes to infectious pneumonia.

### SKILL KEEPER: ELIMINATION HALF-LIFE (SEE CHAPTER 1)

Search "high and low" through drug information resources and you will not find data on the elimination half-life of ethanol! Can you explain why this is the case? The Skill Keeper Answer appears at the end of the chapter.

## D. Treatment of Acute and Chronic Alcoholism

**1. Excessive CNS depression**—Acute ethanol intoxication is managed by maintenance of vital signs and prevention of complications associated with vomiting. Intravenous dextrose is administered to prevent hypoglycemia. Thiamine administration is used to protect against Wernicke-Korsakoff syndrome, and correction of electrolyte imbalances may be required.

**2. Alcohol withdrawal syndrome**—In individuals physically dependent on ethanol, discontinuance can lead to a withdrawal syndrome characterized by insomnia, tremor, anxiety, and, in severe cases, delirium and life-threatening seizures (delirium tremens [DTs]). Peripheral effects include nausea, vomiting, diarrhea, and arrhythmias. The withdrawal syndrome is managed by correction of electrolyte imbalance and administration of thiamine and a sedative-hypnotic. A long-acting benzodiazepine (eg, diazepam or chlordiazepoxide) is usually used unless the patient has compromised liver function, in which case a short-acting benzodiazepine with less complex metabolism (eg, lorazepam) is preferred.

**3. Treatment of alcohol-use disorder**—Alcoholism is a complex sociomedical problem, characterized by a high relapse rate. Several CNS neurotransmitter systems appear to be targets for drugs that reduce the craving for alcohol. The opioid receptor antagonist **naltrexone** has proved to be useful in some patients, presumably through its ability to decrease the effects of endogenous opioid peptides in the brain (see Chapters 31 and 32). A long-acting depot preparation of naltrexone is approved for once-monthly administration. Several antidepressants have been reported to reduce alcohol craving in small studies, but are not approved for this indication. **Acamprosate**, an NMDA glutamate receptor antagonist, is also FDA approved for treatment of alcoholism. The aldehyde dehydrogenase inhibitor **disulfiram** is used adjunctively in some treatment programs. If ethanol is consumed by a patient who has taken disulfiram, acetaldehyde accumulation leads to nausea, headache, flushing, and hypotension (Figure 23–1).

## OTHER ALCOHOLS

### A. Methanol

Methanol (wood alcohol), a constituent of windshield cleaners and "canned heat," is sometimes ingested intentionally. Intoxication causes visual dysfunction, gastrointestinal distress, shortness of breath, loss of consciousness, and coma. Methanol is metabolized to formaldehyde and formic acid, which causes severe acidosis, retinal damage, and blindness. The formation of formaldehyde is reduced by prompt intravenous administration of **fomepizole**, an inhibitor of ADH, or ethanol, which competitively inhibits ADH oxidation of methanol (Figure 23–2).

### B. Ethylene Glycol

Industrial exposure to ethylene glycol (by inhalation or skin absorption) or self-administration (eg, by drinking antifreeze products)